

Project: Milestone 4 Deliverables and Checklist

Hardware for Artificial Intelligence and Machine Learning (HW4AI)

ECE 410/510

Spring 2026

Goal of Milestone 4

Milestone 4 is the complete deliverable package and the basis for your final examination. By June 7, your repository must contain a synthesizable, verified, benchmarked accelerator with a design justification report that an informed reader can use to reproduce your architectural decisions. The final exam draws every question directly from this package.

M4 is final. M4 cannot be revised. Revision tokens do not apply. A No Pass on M4 drops the grade bundle by two letter grades, and No Pass on both M4 and the final examination drops it by four. Plan accordingly.

How to submit

All M4 deliverables are submitted by pushing to your public GitHub repository. There is no Canvas upload.

Expected repository structure for M4

Your repository should contain at minimum the following layout after M4. M1, M2, and M3 paths must still be present. M4 adds `project/m4/` and a top-level `README.md` that points a reader to the M4 deliverables.

```

your-repo/
├── README.md                                ← top-level pointer to M4
├── project/
│   ├── heilmeyer.md
│   ├── m1/                                ← from M1
│   ├── m2/                                ← from M2
│   ├── m3/                                ← from M3
│   └── m4/
│       ├── README.md                      ← catalogs all M4 files
│       ├── rtl/                          ← final source code
│       │   ├── top.sv
│       │   ├── compute_core.sv
│       │   └── interface.sv
│       ├── tb/                            ← final testbenches
│       │   └── tb_top.sv
│       ├── sim/                          ← final simulation outputs
│       │   ├── final_run.log
│       │   └── final_waveform.png
│       ├── synth/                        ← final synthesis results
│       │   ├── config.json
│       │   ├── openlane_run.log
│       │   ├── timing_report.txt
│       │   ├── area_report.txt
│       │   └── power_report.txt
│       └── bench/                        ← hardware vs SW benchmark
│           ├── benchmark.md              ← measured throughput, energy
│           └── benchmark_data.csv        ← raw measurements

```



The `project/m4/` folder is new. The `project/m4/README.md` file is required and must catalog every file in the M4 folder, with a one-line description of contents and a reference to which deliverable (or report section) it supports. The top-level `README.md` must point a reader to the M4 folder and to the design justification report.

Deliverable checklist

Check off each item only when the file is committed and pushed. A checked box that points to a missing or empty file is still Not Yet. The grader runs an automated check on every path; reports, logs, and figures must be committed so the grader can read them without re-running your tools.

Deliverable	GitHub file path
1. README files	
☐ Top-level <code>README.md</code> points to the M4 submission <i>One paragraph minimum. Name the project, link to <code>project/m4/README.md</code> and to the design justification report.</i>	<code>README.md</code>
☐ M4 folder <code>README</code> catalogs every file in <code>project/m4/</code> <i>One line per file: relative path, brief description, which checklist item or report section it supports.</i>	<code>project/m4/README.md</code>
☐ Git tag <code>m4-submission</code> created and pushed <i><code>git tag m4-submission</code> && <code>git push origin m4-submission</code>. The grader uses this tag to locate the graded commit.</i>	(git tag, not a file)
2. Source code	
☐ Final RTL committed: top module, compute core, interface <i>These must be the versions that were synthesized and that produced your final benchmark numbers. If they differ from M3, state the diff in the M4 README.</i>	<code>project/m4/rtl/</code>
☐ Final testbench committed <i>The testbench used to produce the final simulation log. Must be self-contained and runnable from a clean clone.</i>	<code>project/m4/tb/tb_top.sv</code>
☐ Final simulation log committed showing PASS <i>Same PASS/FAIL contract as M2 and M3. Plain text.</i>	<code>project/m4/sim/final_run.log</code>
☐ Final waveform image committed <i>End-to-end transaction. Annotated. PNG or PDF.</i>	<code>project/m4/sim/final_waveform.png</code>

3. Synthesis results

- | | |
|--|---|
| ☐ OpenLane 2 configuration committed
<i>The exact config used to produce the final synthesis run. Clock period, source list, constraints.</i> | <code>project/m4/synth/config.json</code> |
| ☐ OpenLane run log committed
<i>Captured stdout and stderr. Must show the run that produced the timing, area, and power reports below.</i> | <code>project/m4/synth/openlane_run.log</code> |
| ☐ Timing report with critical path and slack
<i>Numerical values for worst negative slack, hold and setup checks. State the clock period your design closes at.</i> | <code>project/m4/synth/timing_report.txt</code> |
| ☐ Area report with cell counts and total area
<i>Total area in μm^2. Cell count by type or by module. The breakdown must be detailed enough for the report to identify the dominant area contributor.</i> | <code>project/m4/synth/area_report.txt</code> |
| ☐ Power report with estimate
<i>Required for M4. If the flow could not produce one, the report must explain what was attempted and why no estimate is available; that explanation will be examined.</i> | <code>project/m4/synth/power_report.txt</code> |

4. Benchmark comparison

- | | |
|---|--|
| ☐ Measured accelerator throughput documented
<i>Same metric as the M1 software baseline: samples/sec, tokens/sec, or FLOP/sec. Method of measurement stated (cycle count from simulation, post-synthesis frequency, or measured if you have FPGA results).</i> | <code>project/m4/bench/benchmark.md</code> |
| ☐ Speedup vs M1 software baseline computed
<i>Speedup = (M1 baseline time) / (M4 accelerator time). State both numbers and the ratio. If your accelerator is slower, say so and explain why.</i> | <code>project/m4/bench/benchmark.md</code> |
| ☐ Energy comparison if available
<i>Optional but valued. Power estimate from synthesis multiplied by accelerator runtime, compared to a published or measured energy number for the SW baseline.</i> | <code>project/m4/bench/benchmark.md</code> |
| ☐ Raw measurement data committed
<i>CSV or equivalent. The numbers behind the summary. The grader and the final examiner may ask where a specific number came from.</i> | <code>project/m4/bench/benchmark_data.csv</code> |
| ☐ Final roofline plot showing where the design sits
<i>Same axes as M1 (FLOP/byte, GFLOP/s, log scale). Plot: target hardware roofline, software baseline point, and the M4 accelerator point. The accelerator point must reflect the measured value, not the M1 hypothetical.</i> | <code>project/m4/bench/roofline_final.png</code> |

5. Design justification report

☐ **Report committed as PDF**

PDF only. Word and markdown are starting points, not deliverables. The final examiner reads the PDF.

`project/m4/report/design_justification.pdf`

☐ **Section: Problem and motivation**

What kernel are you accelerating, why custom hardware. Cite your M1 profiling data with specific numbers.

`project/m4/report/design_justification.pdf`

☐ **Section: Roofline analysis**

Arithmetic intensity of the target kernel. Compute-bound or memory-bound. Where the bottleneck shifts. How the analysis shaped your architecture.

`project/m4/report/design_justification.pdf`

☐ **Section: Precision and data format**

Format used and why. If quantized, the error analysis and verification of acceptability. Reference your M2 precision document.

`project/m4/report/design_justification.pdf`

☐ **Section: Dataflow and architecture**

Dataflow pattern (weight-stationary, output-stationary, input-stationary, no-local-reuse) and why it fits your kernel. Compute engine, memory hierarchy, data path.

`project/m4/report/design_justification.pdf`

☐ **Section: Hardware interface**

Interface implemented, why, effective bandwidth at the target throughput. Whether the design is interface-bound. If so, quantify.

`project/m4/report/design_justification.pdf`

☐ **Section: Verification**

How you verified correctness. Test cases and what they cover. Reference your M2 and M3 testbenches.

`project/m4/report/design_justification.pdf`

☐ **Section: Synthesis results**

Area, timing, power estimate with numbers. Dominant contributor to each. Reference the synthesis reports.

`project/m4/report/design_justification.pdf`

☐ **Section: Benchmark results**

Throughput and/or energy vs software baseline. Explain any gap between measured and theoretical performance.

`project/m4/report/design_justification.pdf`

☐ **Section: What did not work**

Specific. What you attempted that failed, what you learned, what you would do differently. A report without this section is Not Yet.

`project/m4/report/design_justification.pdf`

☐ **Figures referenced in the report committed**

Roofline, block diagram, dataflow diagram, waveforms. Each figure must be referenced from the report text by number.

`project/m4/report/figures/`

☐ **Report is between 2,000 and 5,000 words**

Below 2,000 words, the nine required sections are almost certainly underdeveloped. Above 5,000, the report has drifted from engineering account into literature review.

`project/m4/report/design_justification.pdf`

Common reasons for Not Yet

M4 is the final submission. There is no revision. Each of the following has caused a No Pass in past terms; review every item before you commit and tag.

Report claims something the code does not do

If the report says weight-stationary and the RTL is output-stationary, the final examiner will find it. State what your design actually does. Aspirational descriptions are scored on the actual design, not the aspiration.

Benchmark numbers do not trace to raw data

"5.2x speedup" with no underlying CSV or log is Not Yet. The grader and the final examiner must be able to follow every reported number back to a measurement file in the repository.

Roofline plot uses the M1 hypothetical accelerator point

The M4 roofline must show the measured accelerator point, not the design target from M1. If they are the same, say so; if they differ, the M4 plot must reflect reality.

Power estimate absent without explanation

If OpenLane could not produce a power estimate for your design, document what you tried and what failed. A missing power section with no explanation is Not Yet.

What-did-not-work section is missing or trivial

Every nontrivial design has setbacks. "Everything worked" is rarely true and never an acceptable answer. The section is required. The final examiner will probe what you tried and rejected.

Report sections collapsed into each other

The nine sections must be identifiable. The grader counts them. Combining "Roofline analysis" with "Dataflow and architecture" into a single section is Not Yet, even if the content is good.

Source code and reported design do not match

If the report describes a 16-PE systolic array and the RTL contains an 8-PE design, the discrepancy will be found at the final examination. Update the report or update the code, but submit a coherent package.

Top-level README does not point to the M4 deliverables

A reader landing on the repository home page should be able to find your M4 submission in one click. Top-level `README.md` must do that work.

Filenames or paths do not match the checklist

The grader matches paths exactly. `project/m4/Report/` is not `project/m4/report/`. Use lowercase. Match the names in this checklist character for character.